Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

***Ringkasan*—This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60**

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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[numbers]natbib multicol [bookmarks=true]hyperref

***Ringkasan*—The abstract goes here.**

I. Introduction

This demo file is intended to serve as a “starter file" for the Robotics: Science and Systems conference papers produced under LaTeX using IEEEtran.cls version 1.7a and later.

II. Section

Section text here.

A. Subsection Heading Here

Subsection text here.

1) Subsubsection Heading Here: Subsubsection text here.

III. RSS citations

Please make sure to include `natbib.sty` and to use the `plainnat.bst` bibliography style. `natbib` provides additional citation commands, most usefully `\citet`. For example, rather than the awkward construction

```
\cite{kalman1960new} demonstrated...
```

rendered as "[**?**] demonstrated...," or the inconvenient

```
Kalman \cite{kalman1960new}
demonstrated...
```

rendered as "Kalman [**?**] demonstrated...", one can write

```
\citet{kalman1960new} demonstrated...
```

which renders as "kalman1960new demonstrated..." and is both easy to write and much easier to read.

A. *RSS Hyperlinks*

This year, we would like to use the ability of PDF viewers to interpret hyperlinks, specifically to allow each reference in the bibliography to be a link to an online version of the reference. As an example, if you were to cite "Passive Dynamic Walking" [**?**], the entry in the bibtex would read:

```
@article{McGeer01041990,
  author = {McGeer, Tad},
  title = {\href{http://ijr.sagepub.com/content/9/2/62.abstract}{Passive Dynamic Walking}},
  volume = {9},
  number = {2},
  pages = {62-82},
  year = {1990},
  doi = {10.1177/027836499000900206},
  URL = {http://ijr.sagepub.com/content/9/2/62.abstract},
  eprint = {http://ijr.sagepub.com/content/9/2/62.full.pdf+html},
  journal = {The International Journal of Robotics Research}
}
```

and the entry in the compiled PDF would look like:

[1] Tad McGeer. Passive Dynamic Walking. *The International Journal of Robotics Research*, 9(2):62–82, 1990.

where the title of the article is a link that takes you to the article on IJRR's website.

Linking cited articles will not always be possible, especially for older articles. There are also often several versions of papers online: authors are free to decide what to use as the link destination yet we strongly encourage to link to archival or publisher sites (such as IEEE Xplore or Sage Journals). We encourage all authors to use this feature to the extent possible.

IV. Conclusion

The conclusion goes here.

Acknowledgments